scientific reports

OPEN

A structured multi-day experimental framework integrating green chemistry for the extraction and characterization of Berberine hydrochloride in undergraduate education

Yaqun Liu[1,3,4]✉, Qionglu Huang[2], Zhenxia Zhang[1], Peikui Yang[1], Yongping Huang[1], Jiayi Huang[1], Jie Tan[1], Guohong Xie[1], Jiechun Chen[1], Yicun Chen[3], Chengsong Xie[4] & Yuzhong Zheng[1]✉

To address the challenges of complex workflows, limited student engagement, and insufficient integration of green and interdisciplinary skills in traditional laboratory teaching, this study introduces a multi-day experimental teaching framework with a staged instructional design for the extraction, purification, and identification of berberine hydrochloride (BH) in undergraduate natural product chemistry courses. The framework is implemented as a multi-day practical, pedagogically organized into three instructional stages (extraction, separation and purification, and identification), each of which is aligned with specific learning objectives. This stage-based instructional structure reduces cognitive load and enhances learning outcomes. Green chemistry principles were incorporated into the G-RPWAM (green-reagents, procedures, waste, awareness, methodology) framework to guide solvent selection and procedural design, optimize reagent use, minimize waste, and reduce energy consumption in undergraduate laboratory teaching while fostering environmental awareness. A comprehensive evaluation system, including technical metrics (extraction yield, purity), student performance (experimental skills, data analysis ability, creative thinking, problem-solving ability), and student feedback (content clarity, structured organization, green principles integration, overall effectiveness), was employed to assess the teaching innovation. The results demonstrated that the staged instructional design improved the accessibility and scalability of the experiment, enhanced student satisfaction, and successfully cultivated both sustainability awareness and laboratory skills. This study provides a structured, pedagogy-oriented approach aligned with green chemistry principles for undergraduate natural product chemistry education.

Keywords Natural product chemistry, Berberine hydrochloride, Staged laboratory instruction, Green chemistry, Laboratory teaching reform

Laboratory-based teaching is a cornerstone of chemistry education, providing students with essential hands-on experiences to bridge theoretical knowledge and practical application[1,2]. However, traditional experimental designs often face challenges such as complex workflows, limited student engagement, and insufficient integration of contemporary educational priorities, including sustainability and interdisciplinary skill development[3]. Chemistry education research (CER) has increasingly emphasized evidence-based approaches to laboratory design, with a focus on aligning experimental activities with clearly defined learning outcomes[4]. Studies reported

[1]Guangdong Key Laboratory of Functional Substances and Health Products from Medicinal Edible Resources, Hanshan Normal University, Chaozhou 521041, China. [2]Bioengineering College, Zunyi Medical University, Zhuhai 519090, China. [3]Shantou University Medical College, Shantou 515000, China. [4]Guangdong Taiantang Pharmaceutical Co., Ltd., Shantou 515000, China. ✉email: liuyaqun19900921@126.com; zhengyuzhong@gmail.com

in leading venues such as the Journal of Chemical Education and Chemistry Education Research and Practice suggest that effective laboratory curricula should extend beyond technical skill acquisition to promote conceptual understanding, scientific reasoning, and the progressive development of experimental independence[5,6]. This shift reflects a broader consensus within CER that undergraduate laboratory learning should be intentionally structured to support cumulative skill development rather than isolated procedural training. In the context of natural product chemistry, these issues are compounded by the need to manage intricate extraction and purification processes, which demand both technical precision and a clear understanding of the underlying chemical principles[7].

Berberine hydrochloride (BH), a bioactive alkaloid extracted from *Coptis chinensis* (*C. chinensis*), serves as an ideal model compound for teaching natural product chemistry because of its well-characterized properties and relevance in pharmacological studies[8]. In addition to its pharmacological significance, BH offers distinct pedagogical advantages for structured and sustainability-oriented laboratory instruction in undergraduate natural product chemistry. Its distinctive isoquinoline alkaloid structure, vivid crystalline morphology, and characteristic color changes provide immediate visual feedback and clear correlations between its molecular structure and observable properties, thereby strengthening students' structural-property understanding and molecular identification skills[9]. BH is abundant in widely available *C. chinensis* and can be extracted and purified through simple aqueous acid-base and salting-out recrystallization procedures, using only standard laboratory glassware and low-hazard, green-compatible reagents, without reliance on volatile organic solvents or costly instrumentation. These attributes make BH particularly suitable for large-class teaching while ensuring safety, sustainability, and scalability. Moreover, the workflow and techniques demonstrated with BH are easily transferrable to other alkaloids and natural products, supporting skill migration and cross-course integration across the undergraduate chemistry curriculum. Conventional BH extraction experiments in undergraduate laboratories often lack modularity, fail to incorporate green chemistry principles, and rely heavily on costly instrumentation exemplified by high-performance liquid chromatography (HPLC)[10,11]. These limitations reduce accessibility, scalability, and alignment with modern educational and environmental priorities. In recent years, chemical education research has increasingly emphasized the importance of explicitly integrating educational theory into laboratory experimental design to increase learning efficiency, knowledge transfer, and student engagement[12]. In particular, studies published in core chemistry education journals have highlighted the value of modular and sequential laboratory curricula for supporting cumulative skill development in undergraduate practical chemistry[13,14]. Such designs are especially effective when experiments are intentionally sequenced to revisit core techniques while progressively increasing conceptual depth and learner autonomy. Educational frameworks such as cognitive load theory (CLT), instructional scaffolding, and experiential learning have been widely adopted to rationalize instructional sequencing and task complexity in laboratory teaching. CLT provides a theoretical basis for reducing unnecessary mental burden by segmenting complex experimental workflows into manageable units, thereby optimizing students' working memory during problem solving and skill acquisition[15,16]. Instructional scaffolding supports progressive learning by guiding students from structured tasks toward increasing autonomy, enabling stepwise competency development across experimental stages[17,18]. Moreover, Kolb's experiential learning theory emphasizes the role of concrete experience, reflection, conceptualization, and active experimentation in promoting durable understanding and skill retention[19,20]. Notably, this staged and progressive approach to laboratory instruction aligns closely with the spiral curriculum framework for teaching practical chemistry, in which core experimental skills are revisited across multiple instructional cycles with increasing conceptual sophistication and student independence[21–23]. Within this framework, laboratory experiments function not as isolated training tasks but as interconnected learning experiences that collectively support long-term skill development and conceptual integration across the undergraduate laboratory curriculum. In response to these pedagogical perspectives, this study develops a green, structured, multi-session experimental framework for the extraction, purification, and identification of BH tailored to undergraduate natural product chemistry education. The overall workflow is deliberately divided into three pedagogically structured stages (extraction, separation and purification, and identification), each aligned with clearly defined learning objectives. By applying chunking strategies and scaffolded task progression, the design aims to reduce extraneous cognitive load while enabling students to master one experimental concept or technique before transitioning to the next. Furthermore, experiential learning is embedded within each stage through hands-on operation, real-time observation, and reflective discussion, ensuring that practical execution is closely linked to underlying chemical principles and promoting long-term learning effectiveness.

The integration of sustainability into chemical education has received growing scholarly attention as global environmental challenges demand more responsible scientific practices. Traditional laboratory teaching, while effective in developing technical skills, often involves the use of hazardous reagents, inefficient procedures, and the generation of significant chemical waste[24]. Such practices not only pose safety risks but also conflict with the principles of environmental stewardship, underscoring the need for a shift toward greener and more sustainable approaches in laboratory experimental design. Green chemistry, as a guiding framework, offers a pathway to address these concerns by emphasizing the reduction of hazardous substances, minimizing waste, and improving resource efficiency[25,26]. Its application in laboratory education not only reduces the environmental impact of experiments but also provides an opportunity to cultivate environmental awareness and sustainable thinking among students[27,28]. By integrating these principles into laboratory teaching, educators can align chemistry education with broader sustainability goals, ensuring that students are equipped to contribute to solutions for pressing global issues such as pollution, resource depletion, and climate change[29]. To systematically incorporate green chemistry into experimental teaching, structured methodologies that balance sustainability with educational objectives are needed. This study introduces the G-RPWAM (green-reagents, procedures, waste, awareness, and methodology) framework, an original, practice-oriented model for embedding green chemistry principles into undergraduate laboratory experiments. Through the integration of the twelve principles of green

chemistry with modern pedagogical theories such as CLT and instructional scaffolding[30], G-RPWAM transforms sustainability from a general educational goal into five actionable and assessable dimensions. These dimensions guide the optimization of reagent selection, procedural efficiency, and waste reduction, while systematically cultivating students' environmental awareness and connecting advanced experimental methodologies with low-resource, eco-friendly laboratory designs. By uniting ecological responsibility with modularized learning objectives, the G-RPWAM framework provides a replicable blueprint for developing experiments that are technically rigorous, pedagogically effective, and environmentally friendly.

To evaluate the educational efficacy of this instructional innovation, a comprehensive assessment system was implemented. This system measures technical outcomes, including extraction yield, purification efficiency, and compound purity, alongside student performance metrics (experimental skills, data analysis, creative thinking, and problem-solving abilities). Student feedback was also collected to assess satisfaction with the staged instructional design, clarity of instruction, and integration of green chemistry principles. By combining quantitative and qualitative evaluation methods, this study provides a holistic understanding of the effectiveness of teaching reform. Within this evaluative context, it should be emphasized that the BH experiment presented in this study is intentionally positioned as an early-stage, course-scale undergraduate laboratory structure rather than as an open-ended or research-oriented natural product investigation. Accordingly, the experimental protocol is deliberately designed in a highly structured, step-by-step format, which may be characterized as "cookbook-style" instruction at the surface level. This choice is intentional and pedagogically motivated rather than incidental. For students encountering natural-product extraction and purification for the first time, excessive procedural freedom can impose a substantial extraneous cognitive load, obscure the underlying chemical logic, and increase safety risks in large-enrollment teaching laboratories. Guided by cognitive load theory and instructional scaffolding, the present design stabilizes key experimental variables and operational pathways so that novice learners can focus on mastering transferable core concepts, such as acid-base equilibria, salting-out behavior, crystallization control, and evidence-based compound identification before progressing to more autonomous experimental decision-making. Within a spiral curriculum framework, this structured module therefore functions as a foundational entry point that prepares students for subsequent, more open-ended and research-oriented laboratory experiences rather than replacing problem-solving or exploratory training.

Teaching innovation design and methods

Structured multi-session experimental workflow

To clarify the pedagogical intent of the staged design in this study, the berberine hydrochloride (BH) experiment is organized as a structured multi-session laboratory workflow that accommodates crystallization requirements and standard course scheduling. Here, "staged" denotes a deliberately sequenced and scaffolded instructional design: the workflow is divided into three connected, assessable stages-extraction, separation and purification, and identification-each aligned with specific learning outcomes while contributing to the overall experimental coherence. Guided by cognitive load theory and instructional scaffolding, this sequence is intended to support progressive skill acquisition and minimize extraneous cognitive load in the early stage of a spiral curriculum, preparing students for more open-ended and exploratory laboratory tasks in subsequent courses. Although the operational steps in each stage are explicitly specified, this structured format is not intended to reduce the laboratory experience to procedural replication. Rather, the standardized workflow is designed to foreground causal reasoning by allowing students to focus on why each operation is performed and how specific parameter adjustments influence observable experimental outcomes. By constraining selected variables at this introductory stage, the instructional design enables reliable observation of cause-effect relationships and provides a stable platform for guided questioning, troubleshooting, and reflection, thereby supporting the development of inquiry-oriented thinking within a structured learning environment. The detailed implementation of each stage is described below.

Stage 1 Extraction. This stage focuses on the extraction and preliminary purification of BH via a systematic, stepwise approach. The experimental workflow is designed to enhance students' understanding of chemical extraction principles while ensuring efficiency. (1) Material preparation: A total of 25 g of *C. chinensis* powder was weighed and mixed with 200 mL of 0.2% sulfuric acid solution. The mixture was heated and boiled gently for 40 min, after which the water was replenished as necessary to maintain a constant volume. (2) Secondary handling: The solution from the first extraction was filtered to separate the liquid from the residue. The residue was then subjected to a second extraction using 150 mL of 0.2% sulfuric acid solution, boiled for 20 min, and filtered again. The filtrates from both extractions were combined to maximize yield. (3) Salting out: To purify the extract, lime milk was added to the combined filtrate to adjust the pH to 12, followed by vacuum filtration to remove precipitates. The filtrate was then acidified to pH 2–3 and supplemented with 8% (w/v) sodium chloride. The mixture was stirred until a faint turbidity appeared and then allowed to stand undisturbed to precipitate crude BH (Fig. 1).

Stage 2 Separation and purification. This instructional stage focuses on the systematic separation and purification of BH, highlighting techniques to optimize yield and purity. It strengthens core concepts in separation chemistry while increasing students' experimental skills and analytical accuracy. (1) Crude product treatment: After one week, the BH precipitate was filtered, washed with minimal water until neutral, and vacuum-dried. This duration matches the weekly schedule of the undergraduate laboratory course, allowing crystallization to occur naturally between sessions without extra contact hours. The crude product and filter paper were dissolved in 100 mL of ultrapure water with heating. The filter paper was removed, and the solution was subjected to hot vacuum filtration for further purification. (2) Purification: The filtrate was acidified to pH 2 at 65 °C and then cooled in an ice bath to crystallize yellow precipitates. After crystallization stopped, the crystals were vacuum-filtered, dried at 80 °C for 5 min, and weighed to assess yield and purification efficiency. (3) Recrystallization:

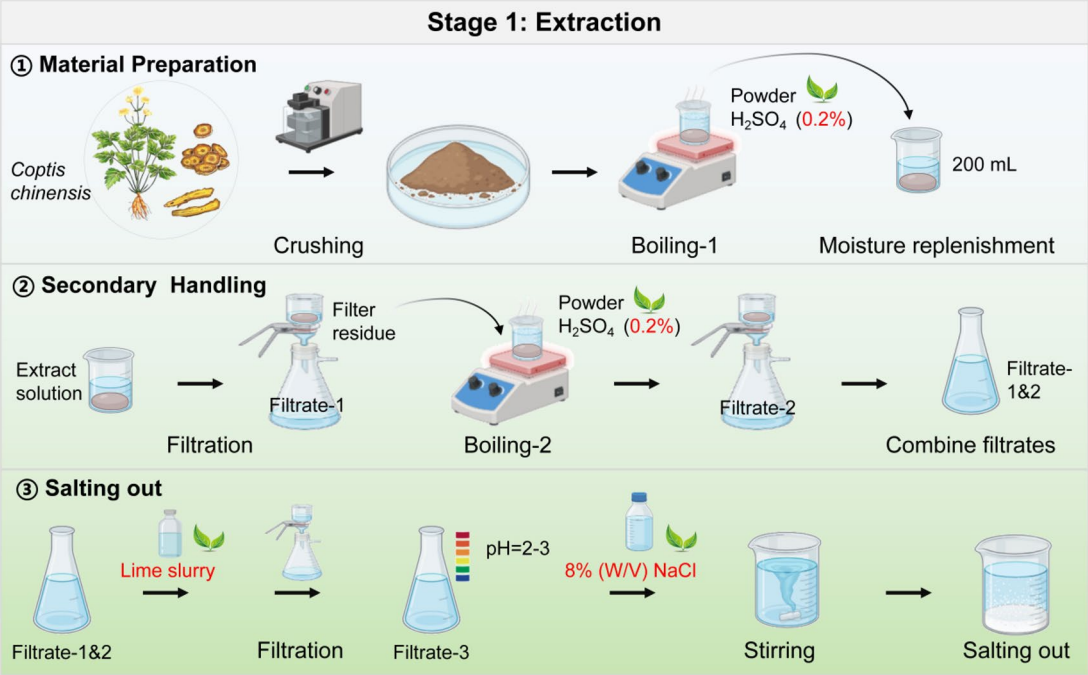

Fig. 1. Structured, stage-based workflow for BH extraction. The diagram shows the stepwise pH and ionic-strength adjustments used in the extraction and the associated phase changes that signal successful transitions. Pedagogical relevance: Students learn to connect each operational adjustment to an expected observation and underlying equilibrium, and instructors can use the checkpoints to scaffold troubleshooting and concept transfer across related separations.

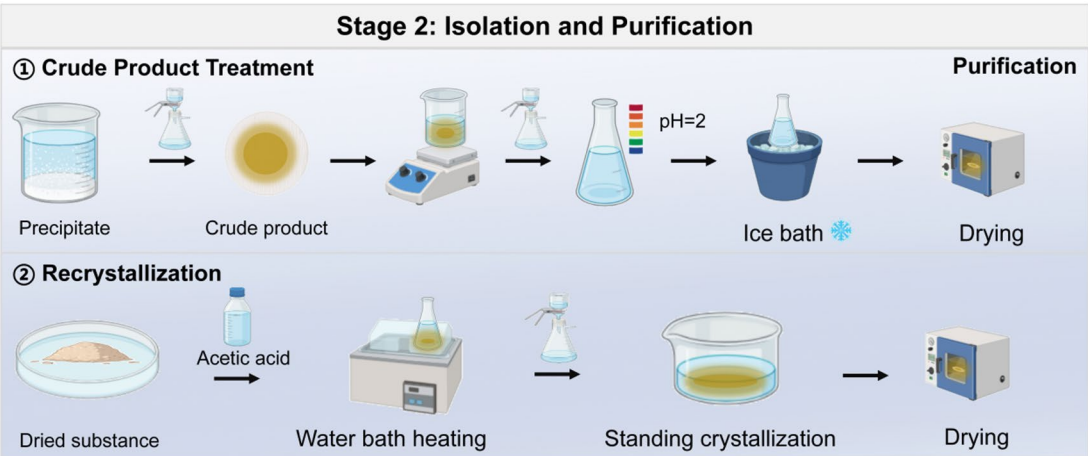

Fig. 2. Systematic workflow for the separation and purification of BH highlighting instructional decision points under green constraints. The figure summarizes the ordered purification steps and the controllable variables (e.g., temperature, solvent volume, and filtration timing) that most strongly affect crystallization and isolation. Pedagogical relevance: Students practice justifying process choices as yield-purity trade-offs, whereas instructors can use the decision points to structure guided inquiry and feedback.

The purified BH was dissolved in acetic acid, heated, and vacuum-filtered. The filtrate was left to crystallize, and the resulting crystals were dried at 60 °C to yield high-purity BH (Fig. 2).

Stage 3 Identification. This instructional stage involves the qualitative and quantitative identification of purified BH via chemical tests, microscopy, and thin-layer chromatography (TLC)[31,32]. It integrates multiple techniques to enhance students' understanding of compound characterization and strengthen experimental skills. (1) Chemical identification: Purified BH was analysed through chemical tests to observe characteristic color changes. The bleaching powder and nitric acid produced a cherry-red color, whereas the reaction with sodium hydroxide in distilled water yielded an orange-red hue, followed by yellow crystalline precipitates

upon the addition of acetone. (2) Microscopic observation: Microscopic analysis of BH crystals and acetone-induced precipitates revealed distinct morphological features. Imaging documented crystallographic structures, offering visual confirmation of purity and structural integrity while reinforcing students' understanding of crystallography. (3) TLC analysis: A silica gel G-CMC TLC plate was used to identify purified BH, with ethanol-dissolved samples and a 1% reference standard. The mobile phase, chloroform-methanol (7:2, v/v), facilitated spot development, and retention factor (*Rf*) values were calculated to assess identity consistency. In this course-scale experiment, the TLC plate was visualized only under UV light. Therefore, TLC provides a rapid check that the purified BH has the same *Rf* as the reference standard and that no additional UV-active spots are observed under the selected conditions. This observation supports identity confirmation and suggests limited UV-active impurities, but it does not by itself constitute definitive proof of chemical purity because non-UV-active impurities, co-migrating components, or trace impurities below the detection limit may not be revealed (Fig. 3).

Incorporating green chemistry principles

The integration of green chemistry principles into this teaching emphasizes procedural optimization and instructional alignment with sustainability concepts rather than exhaustive or industrial-scale environmental performance assessment. This study incorporates green chemistry concepts into traditional BH extraction experiments, optimizing the design to enhance educational outcomes and support more sustainable practices in the undergraduate laboratory context. The G-RPWAM framework (reagents, procedures, waste, awareness, methodology) was established to align the experiment with sustainable natural product extraction practices (Fig. 4).

(1) Reagents (R): Reagent quantities were precisely calculated to minimize waste while achieving teaching objectives. The sulfuric acid concentration was optimized to 0.2%, ensuring efficient extraction of BH from *C. chinensis* without excessive acidity. Sodium chloride addition was controlled at 8% (w/v), with stirring halted at slight turbidity to prevent overuse. Lime has replaced traditional strong bases such as sodium hydroxide, offering a safer, cost-effective, and green chemistry-compatible alternative, which is consistent with the green chemistry principles advocated for ecofriendly reagents.
(2) Procedures (P): The experimental procedure was modified to reduce energy consumption. The extraction process combines low-temperature heating with prolonged simmering to avoid the energy waste associated with high-temperature conditions. During the purification stage, crystallization was achieved by coupling natural cooling with ice-bath techniques, utilizing low-energy cooling methods to facilitate crystal precipitation. These adjustments are designed to lower energy demand in the teaching laboratory setting while maintaining its effectiveness.
(3) Waste Management (W): Waste disposal was carefully addressed in the experimental design. The acidic and alkaline waste solutions generated during the experiment were neutralized prior to discharge, thereby mitigating potential environmental burdens under routine teaching and laboratory disposal practices. Solid waste, represented by filter paper, was dried and collected for centralized processing, reducing the disorderly disposal of experimental waste.
(4) Awareness (A): The teaching process emphasizes the importance of green chemistry principles, guiding students in understanding the environmental considerations underlying experimental design. Students are required to analyse the green chemistry characteristics of each experimental step in their reports and evaluate the potential environmental impacts of the experiment. This approach not only deepens their understanding of experimental principles but also fosters awareness of sustainable development in chemical practices.

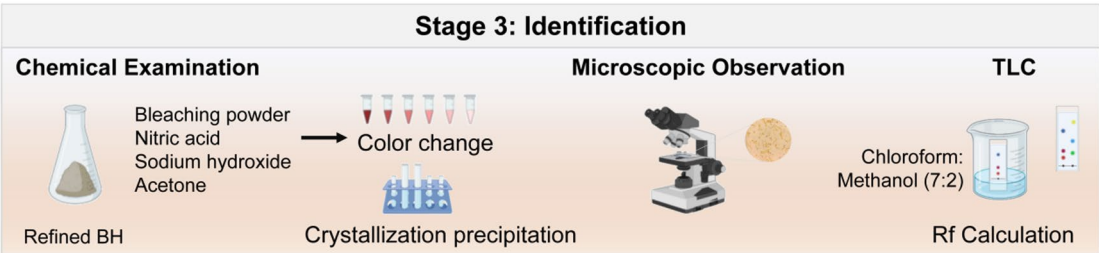

Fig. 3. Evidence-based identity using complementary analytical methods for purified BH. Chemical color reactions, microscopy, and TLC were combined to provide converging evidence for compound identity and an initial impurity screen suitable for an undergraduate teaching laboratory. The diagnostic color reactions provide rapid qualitative cues consistent with BH-type alkaloid behavior, whereas microscopic observation links purification to changes in crystal habit and visible cleanliness. TLC (UV visualization under the selected conditions) supports identity by comigration with a reference standard and screening for additional UV-active components; however, a single aligned spot does not prove chemical purity because non-UV-active impurities, co-migrating species, and trace constituents may remain undetected. Pedagogical relevance: This figure serves as a checklist for stage 3, prompting students to triangulate observations across methods and to articulate both the evidential value and the limitations of each test.

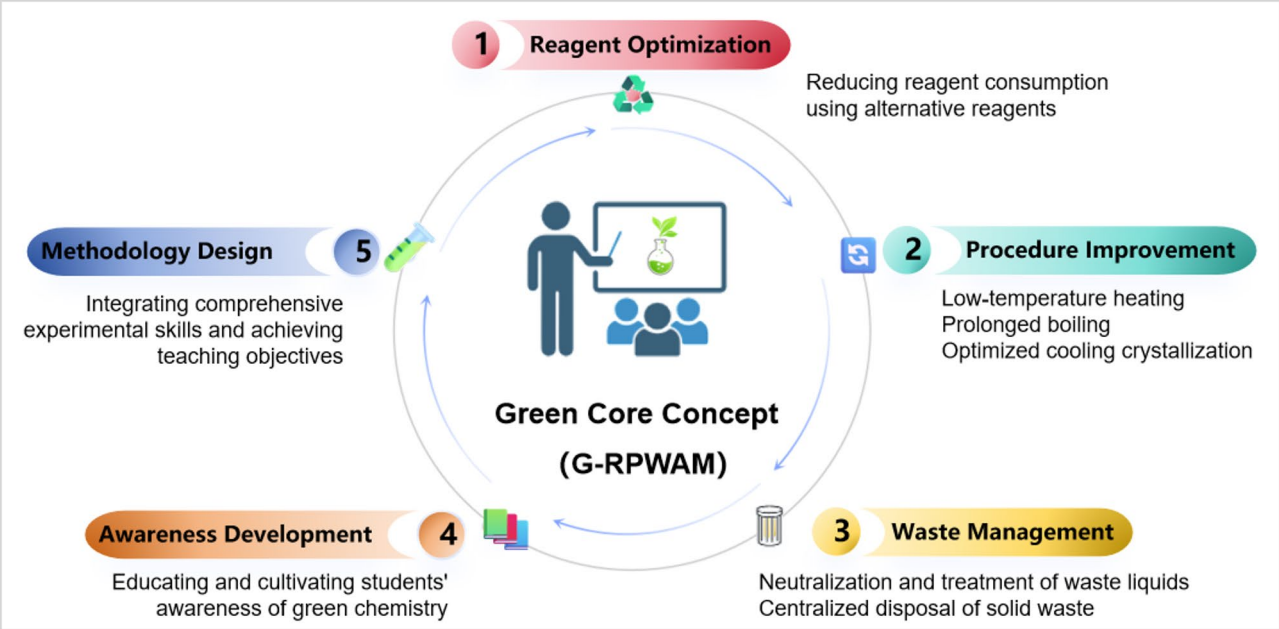

Fig. 4. Framework for integrating the green chemistry principles (G-RPWAM) in experimental teaching. Five dimensions (Reagent, Procedure, Waste, Awareness, Methodology) are embedded within the BH extraction workflow. Pedagogical relevance: Explicitly linking each dimension to a hands-on activity allowing students to translate green chemistry principles into concrete laboratory practices, fostering environmental awareness through direct experimental evidence.

(5) Methodology (M): The experiment integrates foundational techniques, combining salting-out and recrystallization with TLC and microscopic observation, enhancing its comprehensiveness and diversity. Under the guidance of green chemistry principles, the experimental design adheres to the concept of "achieving maximum educational objectives with minimal resource consumption". This balance between technical rigor and sustainability provides students with a holistic learning experience that is consistent with green chemistry principles and aligns with modern environmental and educational standards.

To quantitatively assess the impact of green chemistry integration within the structured experimental workflow at the course scale, explicit green chemistry metrics were incorporated in accordance with established educational-scale sustainability assessment practices. Three categories of parameters were recorded during the experiment: reagent consumption, waste generation, and energy input. All metrics were collected under undergraduate teaching laboratory conditions for comparative and pedagogical purposes rather than for industrial extrapolation.

The actual amounts of reagents used (H_2SO_4, NaCl, and $Ca(OH)_2$) and the total volume of acidic and alkaline waste solutions were measured directly for each experimental batch. On the basis of these data, a simplified E-factor, defined as the ratio of the total mass of chemical waste (excluding aqueous process water) to the mass of purified berberine hydrochloride (BH), was calculated to evaluate the material efficiency:

$$\mathrm{E-factor} = \mathrm{m_{waste}}/\mathrm{m_{product}}$$

The thermal energy input for heating and cooling is calculated from the measured power (P, kW) and operation time (t, h) as follows:

$$\mathrm{E_{thermal}} = \mathrm{P} \times \mathrm{t}$$

Carbon emissions were estimated by converting energy consumption into CO_2 equivalents via the national average electricity emission factor for China ($EFCO_2$=0.785 kg CO_2/kWh):

$$\mathrm{CO_2 emission} = \mathrm{E_{thermal}} \times \mathrm{EFCO_2}$$

To enable meaningful comparisons between experimental designs, carbon emissions were further normalized to the mass of purified BH ($m_{product}$, g) obtained, yielding a unit-product carbon footprint:

$$\mathrm{CO_2 per\ unit\ mass} = (\mathrm{E_{thermal}} \times \mathrm{EFCO_2})/\mathrm{m_{product}}$$

Experimental evaluation system

Student cohort characteristics

This study included three cohorts (C1, C2, and C3, 30 students each) of full-time third-year undergraduates majoring in biotechnology or pharmaceutical engineering, all of whom were enrolled in the compulsory sixth-semester laboratory course Extraction, Separation, and Analysis of Natural Products. Having completed prerequisite courses in general chemistry, organic chemistry, and analytical chemistry, the participants possessed comparable theoretical knowledge and experimental skills. All cohorts shared the same admission year, class composition, and academic background. The course was uniformly delivered by the same lead instructor (Prof. Yuzhong Zheng) with consistent content, laboratory organization, and grading criteria. Laboratory safety, material distribution, and questionnaire/interview coordination were managed by a fixed technician (Dr. Yaqun Liu), who did not participate in grading. Assessment was performed by a stable team applying identical criteria across cohorts, thereby minimizing variability arising from differences in instruction or evaluation.

Evaluation criteria

(1) Experimental outcomes: Key metrics include extraction yield, product purity, and identification accuracy. These indicators reflect the technical success of the experiment and the alignment with theoretical expectations.
(2) Student feedback collection: Student feedback was gathered through two primary methods: anonymous questionnaires and postexperiment interviews.

The questionnaire was developed on the basis of the course objectives and experimental design and covers four core dimensions: content clarity, structured organization, integration of green chemistry principles, and overall effectiveness. Each dimension comprises four items, yielding 16 statements rated on a five-point Likert scale (1 = poor, 5 = excellent)[33]. Representative items included "The explanation of experimental steps enabled independent completion of each stage" (content clarity), "The staged instructional design facilitated progressive acquisition of experimental skills" (structured organization), "The experiment enhanced understanding of minimizing pollution and optimizing reagent use" (green chemistry integration), and "The experimental design improved data analysis and problem-solving abilities" (overall effectiveness). The questionnaire was administered immediately after the experiment and collected anonymously by nonteaching staff (laboratory technicians or assistants). Responses (return rate > 90%) were coded. Reliability and validity were assessed prior to data interpretation. Internal consistency was measured via Cronbach's α, with values ≥ 0.70 considered acceptable[34]. Reverse-scored items were normalized before calculation to avoid bias. Sampling adequacy was examined via the Kaiser-Meyer-Olkin (KMO) statistic, and construct validity was evaluated via Bartlett's test of sphericity. The interview guide covered the clarity of experimental procedures, the timeliness of instructor responses, the value of interactive discussions, and perceptions of integrating green chemistry principles. From each group, five students were selected via stratified random sampling to capture a range of academic performance. The interviews (15–20 min) were conducted by trained nonteaching staff in a quiet, neutral setting and were audio-recorded. The recordings were transcribed verbatim and analysed via Braun and Clarke's six-phase thematic analysis[35]. Two independent coders conducted open coding, organized the codes into subthemes via axial coding, and refined the core themes through selective coding. Coding was managed in NVivo 12 (QSR International). Intercoder reliability was high (Cohen's kappa = 0.84), with discrepancies resolved through discussion. The qualitative results were cross-validated with questionnaire data to enhance reliability.

(3) Instructor evaluation of student performance: Student performance was evaluated across four independent dimensions: experimental skills, data analysis ability, creative thinking, and problem-solving ability. Each dimension was assessed via a 100-point behaviorally anchored rating scale (BARS)[36], with explicit performance descriptors defined for four score ranges (≥ 90, 80–89, 70–79, < 70). The rating system was jointly developed by the course coordinator in collaboration with three senior laboratory instructors and subsequently revised and optimized through a pilot application with a previous cohort of students. Content validity was confirmed through independent review by two peer experts in chemical education. The scoring criteria for each dimension were predefined and based on observable behaviors during laboratory activities. On-site evaluations were conducted strictly according to these predefined standards, and all scores were cross-checked by a laboratory instructor to minimize subjectivity and ensure scoring consistency.

Data analysis

(1) Extraction yield and purification efficiency: The extraction yield and purity are compared with the theoretical values to assess the accuracy and efficiency of the experimental process. The extraction yield was calculated as follows:

$$\text{Extraction yield} = \text{Crude BH weight (M2)} / C.chinensis \text{ powder weight (M1)}.$$

The purification efficiency was determined via the following formula:

$$Purification\ efficiency = Recrystallized\ BH\ weight\ (M3)/Crude\ BH\ weight\ (M2).$$

.

In this study, "purity" refers to a course-scale, method-dependent estimate inferred from recrystallization behavior, microscopic inspection, and TLC cleanliness under UV visualization under the specified conditions rather than an instrument-validated absolute purity value.

(2) Chemical identification: TLC was employed to validate the identity and purity of the extracted compounds, with the *Rf* calculated as:

$$Rf$$

= Distance travelled by the BH (D_{BH})/Distance travelled by the solvent front (D_{SF}).

Color changes and precipitate formation were cross-referenced with standard chemical reactions to confirm the chemical identity of BH.

(3) Correlation analysis: To investigate the relationships between student performance metrics and satisfaction ratings, a comprehensive correlation analysis was performed. The data were normalized to mitigate variability across the three student groups and to account for potential outliers caused by procedural errors. Pearson correlation coefficients were calculated along with their corresponding *p* values and 95% confidence intervals and interpreted with reference to Cohen's effect size criteria (small, medium, and large). Interparameter correlations among the eight performance metrics were analysed to determine potential dependencies or independence among these competencies. Multicollinearity diagnostics were conducted via the variance inflation factor (VIF) and tolerance values, with thresholds set at VIF < 5 and tolerance > 0.2 to ensure the robustness of the correlation estimates. To evaluate the discriminant validity of the defined competency dimensions, confirmatory factor analysis (CFA) was performed via maximum likelihood estimation. Model adequacy was assessed via conventional fit indices, including the comparative fit index (CFI), root mean square error of approximation (RMSEA), and the chi-square to degrees of freedom ratio (χ^2/df). Exploratory factor analysis (EFA) was applied to examine the structural distinctiveness of all competency dimensions.

(4) Advanced statistical analysis: In addition to descriptive statistics, one-way ANOVA was applied to compare extraction yield, purification efficiency, and four dimensions of student performance across the three cohorts (C1-C3). Significant results were further examined via Tukey's HSD post hoc tests. For variables not meeting parametric assumptions, nonparametric Kruskal-Wallis tests were applied to ensure robustness. Microsoft Excel 2019 (Microsoft Corp., Redmond, WA, USA), GraphPad Prism V8.0 (GraphPad Software Inc., La Jolla, CA, USA), SPSS version 23.0 (IBM Inc., Chicago, IL, USA), and R version 4.0.2 (R Foundation, Vienna, Austria) were used. Box plots, heatmaps, radar charts, and Pearson correlation matrices were employed to assess the data distribution, intervariable relationships, and trend dynamics comprehensively.

Safety considerations

All procedures were conducted in compliance with national regulations and the Hanshan Normal University institutional safety code under a unified laboratory risk-management system encompassing hazard identification, probability-severity ranking, control measures, and emergency response planning for each experimental step. The principal hazards included the handling of H_2SO_4, $Ca(OH)_2$ suspensions, NaCl brine, volatile TLC solvents, and elevated-temperature filtration. Acid-alkali-resistant gloves, goggles, and lab coats were used throughout, with all the corrosive or volatile reagents dispensed under fume hoods at controlled rates. Heating and hot filtrations were isolated from combustibles, secured with insulated clamps, splash guards, and face shields, whereas lime milk was freshly prepared and transferred to closed systems to minimize aerosol, inhalation, and clogging risks. pH adjustments and acid-base neutralization were performed under local exhaust ventilation with a portionwise addition of reagent to prevent overheating, and TLC mobile phase preparation and development were completed in enclosed workspaces to avoid vapour exposure. Emergency readiness included trained spill-response protocols, eyewash stations, and neutralization kits, with standardized actions for chemical splashes, bench spills, and glass breakage. Waste streams were classified, clearly labelled, securely sealed, and transferred for certified disposal by the Laboratory Center of Hanshan Normal University. In accordance with green chemistry principles, strong bases were replaced with lime milk where practicable, and unnecessary organic solvent use was minimized to reduce both operator exposure and environmental impact.

Human ethics statement

This study involved course-based educational research with undergraduate students enrolled in a laboratory course at Hanshan Normal University (Chaozhou, China). The Ethics Committee of Hanshan Normal University determined that this study qualified for a waiver of ethics approval under the institutional policies, as it did not involve the collection of sensitive personal data, medical/health-related information, or human biological samples. All participants participated voluntarily and provided informed consent. The study was conducted in accordance with the institutional regulations and ethical guidelines of Hanshan Normal University.

Results and evaluation

Experimental results

Extraction yield and purity

Three student groups were guided through the optimized extraction protocol to isolate BH from 25 g of *C. chinensis* powder (Fig. 5A). The average crude yield of BH across the three groups was 1.95 g, corresponding to a crude extraction rate of 7.81% (Fig. 5B). The results demonstrated high consistency among the groups, with minimal variation observed. Following recrystallization for purification, the refined yield of BH reached

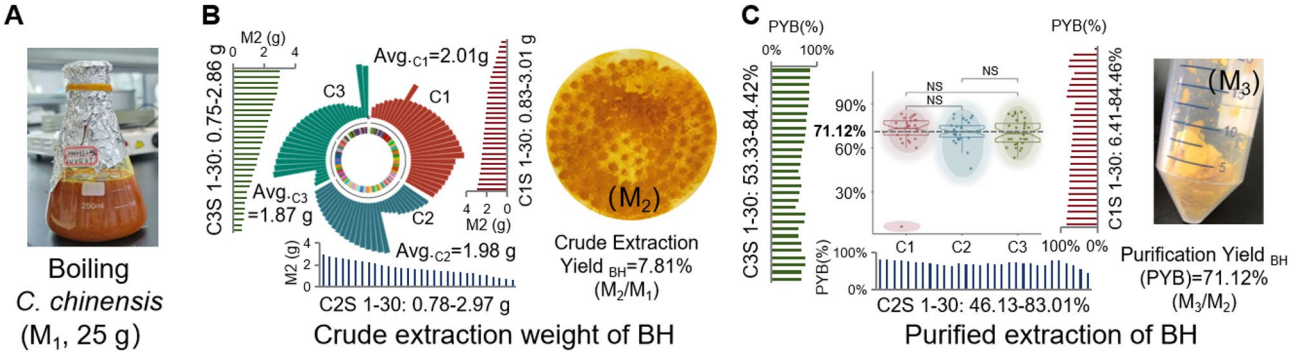

Fig. 5. Extraction and purification outcomes of BH obtained from *C. chinensis* across three student groups. (**A**) Boil extraction of *C. chinensis* powder (M_1 = 25 g) during the initial extraction step. (**B**) Crude extraction weight of BH (M_2). The left panel shows a combined circular distribution and bar plot illustrating individual student results (n = 30 per group). Group C1 is shown in red (mean = 2.01 g, range = 0.83–3.01 g), Group C2 is shown in blue (mean = 1.98 g, range = 0.78–2.97 g), and Group C3 is shown in green (mean = 1.87 g, range = 0.75–2.86 g). The bar plots are ordered by the student index (1–30), with the y-axis representing the crude BH weight. Representative photographs of crude BH following extraction are shown on the right. (**C**) Purified extraction yield of BH after recrystallization. Bar plots display individual student purification yields (students 1–30), whereas the central clustered box-dot plot summarizes the yield distribution across the three groups. Representative photographs of recrystallized BH are shown on the right. Pedagogical relevance: These distributions are used for outcome-based teaching evaluation-helping students learn to interpret variability, identify likely sources of error (e.g., spillage, incomplete filtration, pH overshoot), and relate procedural control to yield/purity; for instructors, the plots support diagnosing which substeps require additional scaffolding or prelab training.

an average of 71.12% across all the groups. While one student in the C1 group experienced a procedural error involving reagent spillage, leading to suboptimal results, the purified extraction rates among the groups remained highly comparable, indicating the robustness of the optimized protocol (Fig. 5C). Relative to the traditional, nonoptimized experiment, the optimized protocol reduced sulfuric acid consumption by 36.4% and sodium chloride consumption by 25.0%, whereas the total volume of acidic/alkaline waste decreased by approximately 41.7% and was fully neutralized prior to discharge. Energy monitoring indicated a 28.9% reduction in electricity use per batch, corresponding to an estimated decrease of 0.43 kg CO_2-equivalent in the carbon footprint. On the basis of the measured waste streams and purified product mass, the optimized experiment exhibited a simplified E-factor of 8.6 ± 0.7 (n = 3 cohorts), corresponding to an approximately 35% reduction relative to the conventional teaching protocol previously used in the same course (E-factor ≈ 13.2). This improvement is attributed primarily to reduced acid consumption, lower salt usage during salting, and decreased aqueous waste generation per unit of BH obtained. Comparative solvent analysis further revealed that the revised protocol relies almost exclusively on water-based systems for extraction and purification, in contrast to reported instructional experiments that employ ethanol, methanol, or multisolvent organic systems. Organic solvent usage was limited to less than 10 mL per batch and restricted to analytical TLC identification, highlighting the enhanced solvent efficiency and improved safety profile of the proposed workflow. These quantitative green metrics substantiate that sustainability gains were achieved without compromising extraction yield, purity, or instructional effectiveness, thereby reinforcing the feasibility of integrating measurable green chemistry principles into undergraduate laboratory teaching.

Chemical identification

The purified BH exhibited characteristic color changes upon interaction with different reagents, serving as qualitative indicators of its chemical properties. When treated with bleaching powder and nitric acid, cherry-red colouration was observed, with the reaction involving bleaching powder yielding a more pronounced red hue. The addition of sodium hydroxide induced an orange colouration, which gradually transitioned into a visible precipitate upon the introduction of acetone (acetone-BH complex) (Fig. 6A). Further analysis via TLC provided UV-based identity confirmation and preliminary impurity screening of the extracted BH. The *Rf* values of the extracted BH matched those of the standard reference compound, and no additional UV-active spots were detected on the plate (Fig. 6B), suggesting the absence of detectable UV-responsive impurities under the applied conditions; however, TLC under UV alone cannot exclude non-UV-active impurities or co-migrating components. The average *Rf* value was 0.71, with individual measurements from the three student groups showing variability around this mean. The *Rf* values from Group C3 demonstrated greater dispersion, whereas an operational error by one participant in Group C2 contributed to localized deviations (Fig. 6C).

Microscopic observations

Microscopic examination of BH in various states revealed distinct structural characteristics. Crude BH exhibited a heterogeneous composition, with clusters of yellow crystalline aggregates resembling BH (Fig. 7A). In contrast, purified BH displayed well-defined yellow crystals, predominantly in the form of elongated needle-like or

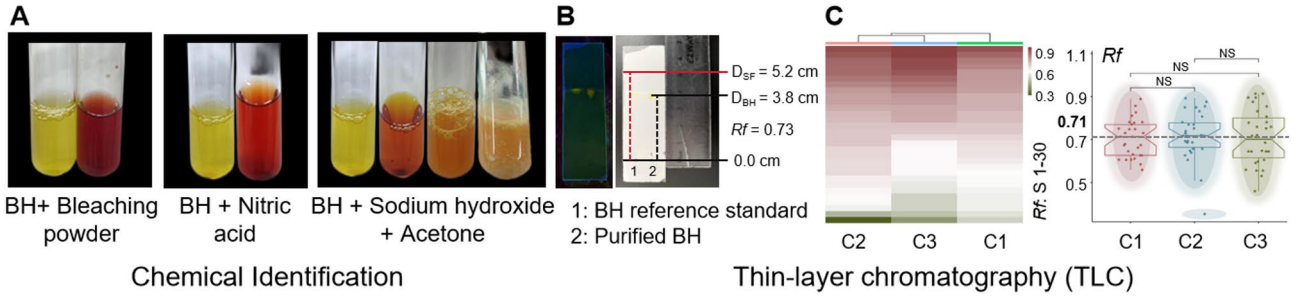

Fig. 6. Chemical identification and purity assessment of purified BH. (**A**) Color reactions of purified BH with selected reagents. (**B**) TLC comparison of the BH reference standard (1) and purified BH (2). The distances between the D_{SF} and D_{BH} spots are marked for *Rf* determination; no extra spots are observed. (**C**) *Rf* analysis: Heatmap showing trends across three student groups and box-dot plot of the *Rf* distribution. Pedagogical relevance: The figure supports instruction on "qualitative evidence + chromatographic evidence" as complementary reasoning, helping students learn to read color reactions, calculate and interpret Rf values against a standard, and recognize purity signals (single spot, consistent Rf) in an undergraduate-accessible manner.

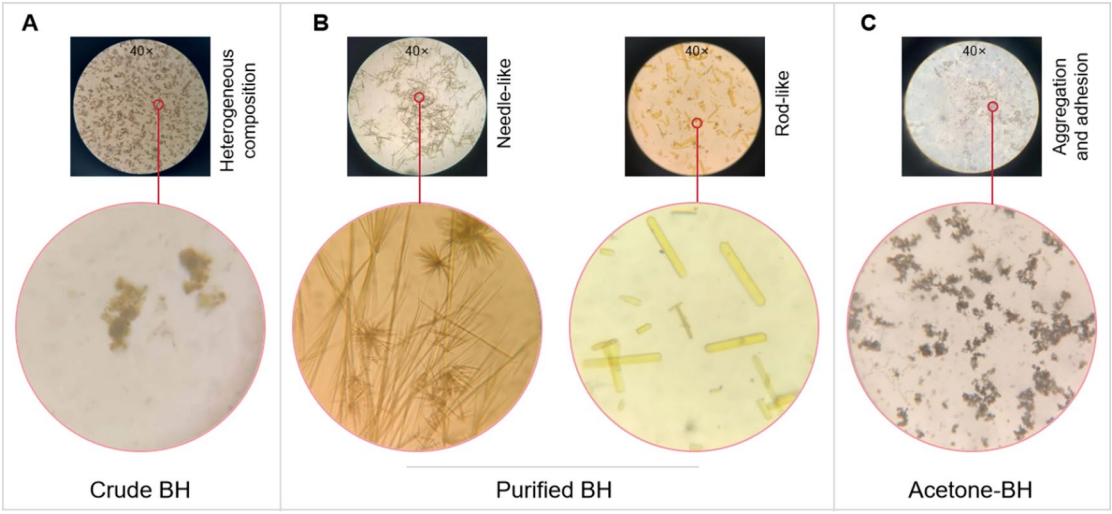

Fig. 7. Microscopic observations of BH in different states. (**A**) Crude BH. (**B**) Purified BH in two morphological forms. (C) Acetone-BH complex. Top row: images acquired under an optical microscope at 40x magnification (black background). Bottom row: corresponding enlarged views of representative crystal morphologies. Pedagogical relevance: These micrographs are used to train students' observation and description skills (crystal habit, aggregation, and impurity cues) and to help instructors prompt comparisons between crude and purified samples, thereby reinforcing how purification steps translate into observable changes in morphology and sample cleanliness.

rod-like structures devoid of impurities (Fig. 7B). The acetone-BH crystals appeared irregular in shape and exhibited aggregation and adhesion (Fig. 7C). Nearly all the students successfully identified the microscopic structures of crude BH and acetone-BH, whereas the majority were able to observe the needle-like crystals of purified BH. Fewer than ten students reported observing the rod-like crystal structures of purified BH. To ensure comprehensive analysis, the students exchanged samples and examined all the observed microscopic structures.

Student performance

Student feedback was collected through anonymous questionnaires and postexperiment interviews, with the evaluation focusing on content clarity, structured organization, green principles integration, and overall effectiveness. Using a 5-point Likert scale, the students rated the instructor's ability to explain and guide complex experimental procedures involving extraction, purification, and identification. The results indicated high levels of satisfaction, with ratings across three student groups ranging from 4.4 to 4.9 and an average score of approximately 4.7 (Fig. 8A). Students particularly appreciated the structured, step-by-step approach, which facilitated their understanding of experimental workflows, and the incorporation of green chemistry principles, which enhanced their awareness of sustainable practices in chemical experimentation. Interactive discussions during the experiments and the instructor's responsiveness to queries were also positively acknowledged.

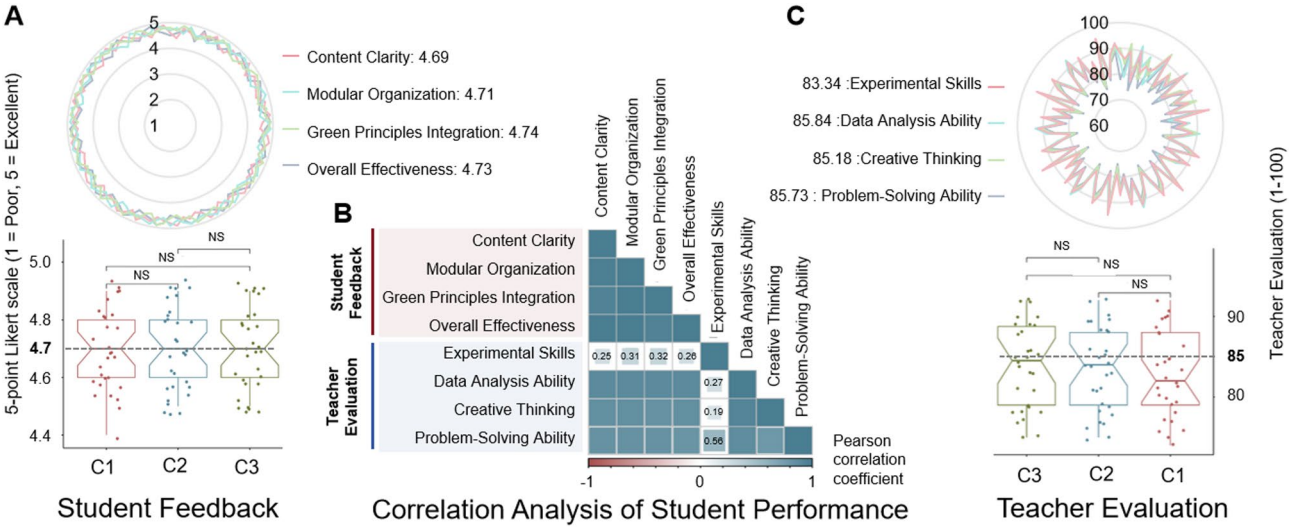

Fig. 8. Comprehensive analysis of student feedback, correlation patterns, and teacher evaluation metrics. (**A**) Student feedback on four instructional dimensions, evaluated with a 5-point Likert scale. The radar chart shows ratings. The box-dot plot shows the score distributions across the three student groups. (**B**) Correlation analysis between student feedback dimensions and instructor evaluations. Visualization includes the strength and direction of correlations among parameters. (**C**) Teacher evaluation scores (100-point scale) for four performance parameters. The radar chart displays the scores; the box-dot plot shows the score distributions across the three student groups. Pedagogical relevance: This figure supports course-level instructional improvement by showing which learning dimensions align with observed performance; it also helps instructors adjust scaffolding (e.g., targeted support for procedural skills) and provides students with transparent criteria linking learning experience, competencies, and assessment.

The questionnaire was subjected to reliability and validity testing. The Cronbach's α coefficients for the four dimensions ranged from 0.973 to 0.987, indicating excellent internal consistency. The KMO value was 0.689, reflecting marginal but acceptable sampling adequacy, and Bartlett's test of sphericity was highly significant (χ^2 = 672.175, df = 6, $p < 0.001$), confirming its suitability for factor analysis. Exploratory factor analysis supported the intended four-factor structure, with clear separation between dimensions.

The teacher assessed student performance across four parameters: experimental skills, data analysis ability, creative thinking, and problem-solving ability. The scores varied significantly, ranging from 64 to 100, with a roughly balanced distribution between high- and low-performing students. While the average experimental skill score was 83.34, the other three parameters consistently exceeded an average of 85.0 across all the groups (Fig. 8C). Correlation analysis between student feedback and instructor evaluations revealed that higher student satisfaction was generally associated with higher instructor ratings. Notably, experimental skills exhibited lower correlations with the other parameters, suggesting that this competency may be relatively independent and less influenced by other factors (Fig. 8B). To examine this trend with greater statistical rigor, VIF and tolerance analyses were conducted, indicating no evidence of severe multicollinearity (all VIFs < 2.5, tolerance > 0.5). Partial correlation analyses controlling for GPA, grade level, and satisfaction ratings further reduced the associations between experimental skills and the other dimensions. Exploratory factor analysis revealed low cross-loadings for experimental skills (< 0.35), and confirmatory factor analysis supported its relative independence (CFI = 0.961, RMSEA = 0.043). Given the observational nature of these findings, this independence should be interpreted as an exploratory observation rather than a strong causal inference.

Discussion

The structured, multi-session experimental workflow integrating green chemistry principles provided quantifiable improvements in pedagogical effectiveness and procedural sustainability indicators within undergraduate natural product extraction and characterization[27,37]. The stage-based instructional structure, implemented over multiple laboratory sessions, enhanced procedural clarity and reduced the cognitive load by dividing the experimental process into discrete units targeting specific competencies. In contrast to conventional phytochemistry or alkaloid extraction designs, which are typically restricted to single-step optimization or rely on professional-scale instrumentation without structured pedagogical segmentation[38,39], the structured framework combines extraction, purification, and identification into a coherent workflow explicitly aligned with the G-RPWAM sustainability dimensions. Embedding reagent optimization, waste minimization, and energy-saving strategies within each stage ensured that environmental metrics, such as reductions in acid use, waste volume, and the carbon footprint, were directly measurable alongside student performance indicators. This integration supports stepwise technical skill development while systematically reinforcing underlying chemical theory, thereby linking learning outcomes to verifiable environmental gains.

Quantitative and qualitative evaluations confirmed that the structured experimental workflow substantially improved clarity, organization, and student engagement in undergraduate natural product laboratory teaching. In accordance with Kolb's experiential learning cycle, each stage combines defined learning objectives with hands-on tasks and incorporates scaffolding and chunking strategies to lower extraneous cognitive load, enabling students to master one stage before progressing to the next. By segmenting extraction, purification, and identification into discrete steps-such as salting-out, controlled crystallization, and TLC analysis-the design strengthened both technical skills and conceptual understanding without increasing procedural complexity. Stable extraction yield and purification efficiency across the three cohorts demonstrated protocol robustness under typical teaching-lab variability, whereas high satisfaction scores for "content clarity" and "structured organization" reflected its pedagogical value. The integration of green chemistry principles within each stage links procedural segmentation to measurable environmental benefits, ensuring that methodological rigor is matched by sustainability outcomes[40]. Future iterations could incorporate advanced yet low-cost upgrades such as two-dimensional TLC (2D-TLC) for orthogonal compound verification or the substitution of traditional eluents with greener alternatives, further expanding the methodological scope and reinforcing the evidence chain. The structured progression also facilitated engagement with advanced characterization techniques in a supportive framework, promoting the transfer of skills to related laboratory contexts[41].

Integrating green chemistry principles into the structured workflow directly addresses the environmental dimension of laboratory teaching, moving beyond the common emphasis on technical skills alone[42]. Unlike purely qualitative sustainability descriptions, the present study embeds quantitative green chemistry metrics (reagent reduction ratios, simplified E-factors, solvent profiling, and unit-product carbon footprints) directly into the experimental evaluation system, enabling a transparent comparison with traditional teaching experiments and strengthening the evidentiary basis for sustainability-oriented laboratory reform. The incorporation of the G-RPWAM sustainability framework supports systematic minimization of waste generation, hazard potential, and resource consumption and-distinct from widely adopted macro-level frameworks such as the Twelve Principles of Green Chemistry-offers direct operability by translating each dimension into an actionable teaching unit that unites environmental objectives with specific skill-building activities. This micro- to-macro alignment ensures that sustainability is embedded at the procedural level rather than remaining an abstract guideline, allowing course scale quantification of resource use and waste handling as instructional evidence supporting alignment with green chemistry principles. Solvent volumes and energy inputs were deliberately optimized at each stage without compromising extraction yields or analytical accuracy[43]. A notable illustration is the one-week precipitation period in the purification stage, deliberately synchronized with the standard weekly schedule of undergraduate laboratories to allow crystallization under ambient conditions without the energy demands of rapid cooling. This scheduling strategy ensures the formation of uniform crystals that facilitate subsequent filtration and purification, exemplifying the integration of logistical efficiency, green chemistry, and reproducibility[44]. These adjustments resulted in measurable reductions in waste volume and toxicity indices compared with the baseline values from traditional protocols. Notably, all the quantitative green chemistry metrics were derived at the undergraduate teaching laboratory scale, and the carbon footprint values were estimated via a simplified life-cycle assessment model for in-course comparative purposes only, without direct extrapolation to industrial production. Moreover, while short-term improvements in students' environmental awareness are supported by quantitative evidence, the persistence of such effects over the long term warrants verification through future longitudinal studies[45]. By embedding sustainability evaluations into the experimental workflow, students can directly connect chemical processes to their environmental consequences, thereby reinforcing awareness of green chemistry in practical contexts[42]. The integration of sustainability into teaching design thus advances dual objectives: technical proficiency and environmental responsibility.

The evaluation of student performance across experimental skills, data analysis, creative thinking, and problem-solving abilities provided a multidimensional assessment of learning outcomes. While all the parameters presented high average scores, the experimental skills demonstrated a weaker correlation with data analysis and creative thinking, indicating that procedural proficiency may develop independently from analytical and creative abilities under structured laboratory conditions[46]. This divergence, supported by variance inflation factor and confirmatory factor analyses, suggests context-specific independence rather than universal separation, highlighting a potential need for differentiated instructional strategies[47]. Instead of merely reiterating performance statistics, these findings were interpreted as evidence that skill-focused stages may require additional scaffolding, such as pre-lab simulations or targeted micro-assessments, to bridge gaps for students facing procedural challenges[48,49]. The robustness of the feedback and correlation patterns underscores the reliability of this observation. From a pedagogical perspective, aligning hands-on technical training more closely with stages that cultivate higher-order reasoning could enhance knowledge transfer, ensuring that procedural mastery also reinforces analytical and creative competencies[50]. Importantly, the analytical framework adopted in this study is fundamentally descriptive and exploratory. The applied statistical analyses are intended to characterize patterns, trends, and associations observed within the implemented instructional context rather than to test predefined hypotheses or establish causal relationships between the structured green experimental framework and student learning outcomes.

The implementation of this experimental teaching design reveals specific challenges and opportunities in advancing undergraduate natural product chemistry education. Large class sizes increase safety risks when handling hazardous chemicals or performing high-temperature operations[51], whereas dependence on costly, limited-access instruments imposes financial and logistical burdens on departments[52]. The adoption of yield analysis, microscopy, and TLC as low-cost, high-value alternatives reduces reliance on such instrumentation without compromising educational quality, ensuring that students acquire essential laboratory competencies. Notably, the TLC assessment in this study was based solely on UV visualization; therefore, a single co-migrating UV-active spot supports identity consistency under the chosen conditions but cannot exclude co-migrating

or non-UV-active impurities. Accordingly, "purity" is reported as an apparent, course-scale indicator rather than an instrument-validated purity value (e.g., HPLC, liquid chromatography-mass spectrometry (LC-MS), or nuclear magnetic resonance (NMR)). Nonetheless, variability in procedural skills indicates that some learners may require additional scaffolding. Supplementary resources, including prelab tutorials and step-by-step instructional videos, could improve technical proficiency. Virtual laboratory platforms offer additional low-cost pathways to deliver and scale stage-based workflows[53]. Although green chemistry integration has been well-received, a subset of students struggle to connect specific procedural modifications with broader environmental impacts; embedding targeted case studies or real-world scenarios in teaching materials may strengthen this link[29]. Interpretations of student performance and competency development should be approached with appropriate caution. As the study does not incorporate randomized group assignment, control conditions, or formal pre- and post-intervention comparisons, references to improvements in experimental skills, data analysis capabilities, or higher-order cognitive competencies reflect observed performance distributions and self-reported learning experiences rather than demonstrable causal effects attributable to the instructional design. Furthermore, refining feedback instruments to maintain clarity and reduce redundancy would enhance psychometric robustness and support broader application. Extending the current evaluation framework to longitudinal studies could provide deeper insight into the persistence of skill and awareness gains, informing future refinements. The integration of AI-assisted analytics into structured experimental design could further increase learning efficiency[54]. Addressing these targeted improvements while retaining the strengths of the present design could advance both the quality and sustainability of laboratory education. Accordingly, the principal contribution of this work lies in systematically documenting the design, implementation, and evaluative characteristics of a structured green laboratory framework within undergraduate natural product chemistry education. The findings provide an empirical basis for understanding its pedagogical features and practical applicability. Further research employing controlled or longitudinal study designs would be necessary to rigorously assess causal impacts on learning gains and long-term educational outcomes. This experiment is intentionally positioned as a course-scale skills platform rather than a research-grade natural-product separation project. In this study, the adoption of a structured, step-by-step ("cookbook-style") protocol reflects a deliberate instructional boundary rather than a constraint on pedagogical intent. For early-stage undergraduate laboratory instruction, the primary objective is the consistent acquisition of foundational experimental competencies under conditions that are safe, reproducible, and scalable. Limiting procedural variability at this stage reduces extraneous trial-and-error and enables instructional emphasis on conceptual understanding, interpretation of experimental observations, and guided troubleshooting. Within a spiral curriculum framework, these competencies are subsequently reinforced in later laboratory teaching through progressively reduced scaffolding and increased student autonomy. The workflow therefore stabilizes key variables and prioritizes transferable core concepts (extraction equilibria, salting-out, crystallization control, and evidence-based identification) over open-ended strategy design or exhaustive method benchmarking. This boundary is a pedagogical choice aligned with cognitive-load management and spiral-curriculum progression, preparing students for later stages that explicitly require greater autonomy and method comparison.

Limitations

The present work is intended to document a feasible teaching implementation and its observable classroom/laboratory outcomes; thus, the scope of inference is necessarily bounded. The participants came from a limited instructional setting (three cohorts within the same course), so transferability to other student populations and curricula should be made with caution. In addition, because the teaching was carried out under authentic course conditions without random allocation and without a parallel control condition, the reported outcome patterns primarily reflect instructional implementation and learning observation rather than a strict attribution of effects to a single factor. Moreover, the outcome evidence was derived mainly from end-of-course performance/ratings and perception-based feedback, without a structured pre-post design; therefore, causal claims should be avoided. The evaluation window was confined to the course cycle, and longer-term retention of operational skills and sustainability awareness were not examined. Therefore, the findings should be read as practice-oriented evidence supporting the educational value of structured green design, not as conclusive causal proof, and should not be overextended beyond the stated context. Finally, the green metrics reported here are teaching-scale benchmarks intended for within-course comparison and student awareness building, and they should not be extrapolated to process-level or industrial environmental performance.

Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Received: 18 October 2025; Accepted: 3 February 2026
Published online: 10 February 2026

References

1. Egambaram, O. et al. The future of laboratory chemistry learning and teaching must be Accessible. *J. Chem. Educ.* **99** (12), 3814–3821 (2022).
2. Guntero, V. A., Mancini, P. M. E. & Kneeteman, M. Introducing organic chemistry students to the extraction of natural products found in vegetal Species. *World J. Chem. Educ.* **4** (5), 142–147 (2017).
3. Onuoha, C. P., Garner, M. P. & Fenn, N. E. R. Student perceptions of co-curricular activities on pharmacy education: A review. *Curr. Pharm. Teach. Learn.* **13** (5), 576–584 (2021).
4. Cooper, M. M. & Stowe, R. L. Chemistry education Research-From personal empiricism to Evidence, Theory, and informed Practice. *Chem. Rev.* **118** (12), 6053–6087 (2018).

5. Wu, M. M. & Yezierski, E. J. Investigating Teacher-Teacher feedback: Uncovering useful Socio-pedagogical norms for Reform-Based chemistry Instruction. *J. Chem. Educ.* **100** (11), 4224–4236 (2023).
6. Nugraheni, A. R. E. & Srisawasdi, N. Development of pre-service chemistry teachers' knowledge of technological integration in inquiry-based learning to promote chemistry core competencies. *Chem. Educ. Res. Pract.* **26** (2), 398–419 (2025).
7. Zhang, L. et al. The strategies and techniques of drug discovery from natural products. *Pharmacol. Ther.* **216**, 107686 (2020).
8. Zhao, J. et al. Berberine hydrochloride ameliorates PM2.5-induced pulmonary fibrosis in mice through inhibiting oxidative stress and inflammatory. *Chem. Biol. Interact.* **386**, 110731 (2023).
9. Xiao, Y. et al. Berberine hydrochloride enhances innate immunity to protect against pathogen infection via p38 MAPK pathway. *Front. Immunol.* **16**, 1536143 (2025).
10. Yang, Y. et al. Determination of alkaloid contents in various tissues of Coptis chinensis Franch. By reversed Phase-High performance liquid chromatography and ultraviolet Spectrophotometry. *J. Chromatogr. Sci.* **55** (5), 556–563 (2017).
11. Wang, Y. et al. Simultaneous determination of eight active compounds in Baitouweng decotion and its single Herbs. *J. Chromatogr. Sci.* **57** (6), 502–510 (2019).
12. Jakubčinová, J., Feszterová, M. & Silliková, V. Active learning in the extraction of organic compounds: A study of undergraduate chemistry Students. *Educ. Sci.* **14** (10), 1051 (2024).
13. Wackerly, J. W., Dunne, J. F. & Shriver, J. A. Multistep synthesis of diphenhydramine: A modular laboratory for First-Semester organic chemistry Students. *J. Chem. Educ.* **102** (7), 2859–2865 (2025).
14. Fleischer, T. et al. Digital sequential scaffolding during experimentation in chemistry Education-Scrutinizing influences and effects on Learning. *Educ. Sci.* **13** (8), 811 (2023).
15. Young, J. Q. et al. Cognitive load theory: implications for medical education: AMEE guide 86. *Med. Teach.* **36** (5), 371–384 (2014).
16. Sweller, J. Cognitive load during problem solving: effects on learning. *Cogn. Sci.* **12** (2), 257–285 (1988).
17. Yin, D. S. & Yin, X. Scaffolding learning: from specific to generic with large Language models. *PloS One*. **19** (9), e310409 (2024).
18. Wood, D., Bruner, J. S. & Ross, G. The role of tutoring in problem solving. *J. Child Psychol. Psychiatry Allied Discip.* **17** (2), 89–100 (1976).
19. Long, E. M. & Gummelt, G. Experiential service learning: Building skills and sensitivity with kolb's learning theory. *Gerontol. Geriatr. Educ.* **41** (2), 219–232 (2020).
20. Kolb, D. A. *Experiential Learning: Experience as the Source of Learning and Development*. (Prentice-Hall, 1984).
21. Worrall, A. F. et al. University teaching of mass spectrometry as a key practical technique within the context of a fully integrated, spiral curriculum. *Rapid Commun. Mass Spectrom. RCM.* **30**, e9851 (2024).
22. Fauzi Ah, L. et al. Modified spiral organic curriculum on organic chemistry courses for chemistry education undergraduate students. *AIP Conf. Proc.* **1911**(1), (2017).
23. Schmidkunz, H. & Büttner, D. *Teaching Chemistry According To a Spiral curriculum* (European Journal of Science Education, 1986).
24. Freese, T. et al. The relevance of sustainable laboratory practices. *RSC Sustain.* **2** (5), 1300–1336 (2024).
25. de Marco, B. A. et al. Evolution of green chemistry and its multidimensional impacts: A review. *Saudi Pharm. J.* **27** (1), 1–8 (2019).
26. Tobiszewski, M., Mechlińska, A. & Namieśnik, J. Green analytical chemistry-theory and practice. *Chem. Soc. Rev.* **39** (8), 2869–2878 (2010).
27. Etzkorn, F. A. & Ferguson, J. L. Integrating green chemistry into chemistry Education. *Angew Chem. Int. Ed. Engl.* **62** (2), e202209768 (2023).
28. de la Parra, J. et al. *Crafting a More Environmentally Benign Extraction and Analysis of Pharmaceutical Precursors from a Medicinal Plant: A Student-Led Innovation* (ChemRxiv Preprint, 2019).
29. Plotka-Wasylka, J. et al. Green chemistry in higher education: State of the Art, Challenges, and future Trends. *ChemSusChem* **11** (17), 2845–2858 (2018).
30. Krasnodębski, M. The bumpy road to sustainability: Reassessing the history of the twelve principles of green chemistry. *Stud. Hist. Philos. Sci.* **103**, 85–94 (2024).
31. Lu, J. et al. Separation of Berberine hydrochloride and Tetrahydropalmatine and their quantitative analysis with thin layer chromatography involved with ionic Liquids. *J. Anal. Methods Chem.* **2015**, 642401 (2015).
32. Zhen, H. S., Chen, Y. & Zhong, X. J. Quality standards of Huanglian Jiedu pills. Zhongguo Zhong Yao Za zhi = Zhongguo Zhongyao Zazhi = China. *J. Chin. Materia Med.* **19** (7), 414–447 (1994).
33. Jebb, A. T., Ng, V. & Tay, L. A review of key likert scale development advances: 1995–2019. *Front. Psychol.* **12**, 637547 (2021).
34. Korrel, M. et al. Framework for training in minimally invasive pancreatic surgery: An international Delphi consensus Study. *J. Am. Coll. Surg.* **235** (3), 383–390 (2022).
35. Braun, V. & Clarke, V. Using thematic analysis in psychology. *Qual. Res. Psychol.* **3** (2), 77–101 (2006).
36. Lynch, B. L. A behaviorally anchored rating scale for the evaluation of student performance. *Am. J. Med. Technol.* **43** (1), 54–63 (1977).
37. Miladinović, S. M. Green analytical chemistry: Integrating sustainability into undergraduate education. *Anal. Bioanal. Chem.* **417** (4), 665–673 (2025).
38. Liu, Z. et al. Comprehensive undergraduate experiment: Determination of phenylurea herbicide in tea by magnetic solid-phase extraction-high performance liquid chromatography. *Se pu = Chin. J. Chromatogr.* **42** (11), 1094–1100 (2024).
39. Ho, C. C. et al. Employing pressurized hot water extraction (PHWE) to explore natural products chemistry in the undergraduate Laboratory. *J. Vis. Exp.* **7** (141), 10–3791 (2018).
40. Murcia, J. E. et al. Risk assessment and green chemistry applied to waste generated in university laboratories. *Heliyon* **9** (5), e15900 (2023).
41. Heni, M. et al. Focused didactic training for skills lab student tutors - which techniques are considered helpful?. *GMS Z. Fur Medizinische Ausbildung*. **29** (3), Doc41 (2012).
42. Chen, M., Jeronen, E. & Wang, A. What Lies behind teaching and learning green chemistry to promote sustainability education? A literature review. *Int. J. Environ. Res. Public. Health.* **17**(21), 7876, (2020)
43. Sahoo, T. et al. Green solvent: green shadow on chemical Synthesis. *Curr. Org. Synth.* **17** (6), 426–439 (2020).
44. Rahman, M. U. et al. Green analytical techniques for impurity determination in Pharmaceuticals. *Electrophoresis*. **47**(1):87–105 (2026).
45. Collins, J. et al. Impact of an international service learning experience in India for DPT students: Short- and Long-Term Benefits. *J. Allied Health*. **48** (1), 22–30 (2019).
46. Bowling, A. M. & Underwood, P. W. Effect of simulation on knowledge, self-confidence, and skill performance in the USA: A quasi-experimental study. *Nurs. Health Sci.* **18** (3), 292–298 (2016).
47. Xu, B. et al. How does mathematical literacy affect creative thinking? Independent effects and differential impacts across proficiency groups. *Acta. Psychol.* **260**, 105509 (2025).
48. Smerdel, S. & Zejnilagić Hajrić, M. Demographic characteristics of chemistry teachers in Croatia affecting the use of Pre-laboratory activities in the Classroom. *Acta Chim. Slov.* **67** (2), 435–444 (2020).
49. Coleman, A. B. et al. Imagining, designing, and interpreting experiments: Using quantitative assessment to improve instruction in scientific reasoning. *Biochem. Mol. Biol. Educ.: Bimon. Public. Int. Union Biochem. Mol. Biol.* **51** (3), 286–301 (2023).
50. Basey, J. M. et al. An evaluation of two hands-on lab styles for plant biodiversity in undergraduate biology. *CBE Life Sci. Educ.* **13** (3), 493–503 (2014).

51. Fatemi, F., Dehdashti, A. & Jannati, M. Implementation of chemical Health, Safety, and environmental risk assessment in laboratories: A Case-Series Study. *Front. Public. Health*. **10**, 898826 (2022).
52. Cooley, C. Z. et al. Design and implementation of a low-cost, tabletop MRI scanner for education and research prototyping. *J. Magn. Reson.* **310**, 106625 (2020).
53. Reyes, R. L. et al. Enhancing experiential science learning with virtual labs: A narrative account of merits, challenges, and implementation strategies. *J. Comput. Assist. Learn.* **40** (6), 3167–3186 (2024).
54. Chiu, W. Pedagogy of emerging technologies in chemical education during the era of digitalization and artificial intelligence: A systematic review. *Educ. Sci.* **11**, 709 (2021).

Author contributions

Yaqun Liu: Conceptualization, methodology, supervision, writing-original draft preparation, writing-review and editing. Qionglu Huang: Experimental implementation. Zhenxia Zhang: Experimental design, data analysis, visualization. Peikui Yang: Development of the evaluation system, statistical analysis. Yongping Huang: Experimental implementation. Jiayi Huang: Experimental trials, collection of student feedback. Jie Tan: Experimental trials, collection of student feedback. Guohong Xie: Experimental trials, collection of student feedback. Jiechun Chen: Experimental trials, collection of student feedback. Yicun Chen: Analysis of student performance. Chengsong Xie: Microscopy observations, manuscript review. Yuzhong Zheng: Supervision, theoretical framework development, writing-review and editing. All the authors discussed the results and contributed to the final manuscript.

Funding

This work was supported by the Guangdong Key Laboratory of Functional Substances and Health Products from Medicinal Edible Resources (Grant Number 2021B1212040015), the Guangdong Provincial Department of Education Project (Grant Number 2025KQNCX043), the Hanshan Normal University Quality Engineering Program (Grant Number HSJG-SZ252187, E25070 and e25070), the Third Batch Professor Startup Project of Hanshan Normal University (Grant Number QD2023312), the Hanshan Normal University Scientific Research Mentor Support Program (Grant Number XBF202302), and the Chaozhou Science and Technology Bureau Science and Technology Project (Grant Number 202404GY06).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Y.L. or Y.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026